Saving the Results

Different Formats

While we run various scans, we should always save the results. We can use these later to examine the differences between the different scanning methods we have used. `Nmap` can save the results in 3 different formats.

- Normal output (`oN`) with the `.nmap` file extension
- Grepable output (`oG`) with the `.gnmap` file extension
- XML output (`oX`) with the `.xml` file extension

We can also specify the option (`-oA`) to save the results in all formats. The command could look like this:

Saving the Results

```
pecna@htb[/htb]$ sudo nmap 10.129.2.28 -p- -oA target

Starting Nmap 7.80 ( https://nmap.org ) at 2020-06-16 12:14
CEST
Nmap scan report for 10.129.2.28
Host is up (0.0091s latency).
Not shown: 65525 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh
25/tcp    open  smtp
80/tcp    open  http
MAC Address: DE:AD:00:00:BE:EF (Intel Corporate)

Nmap done: 1 IP address (1 host up) scanned in 10.22 second
s
```

Scanning Options	Description
`10.129.2.28`	Scans the specified target.
`-p-`	Scans all ports.

Scanning Options	Description
`-oA target`	Saves the results in all formats, starting the name of each file with 'target'.

If no full path is given, the results will be stored in the directory we are currently in. Next, we look at the different formats `Nmap` has created for us.

```
pecna@htb[/htb]$ ls

target.gnmap target.xml  target.nmap
```

Normal Output

```
pecna@htb[/htb]$ cat target.nmap# Nmap 7.80 scan initiated
Tue Jun 16 12:14:53 2020 as: nmap -p- -oA target 10.129.2.2
8Nmap scan report for 10.129.2.28
Host is up (0.053s latency).
Not shown: 4 closed ports
PORT   STATE SERVICE
22/tcp open  ssh
25/tcp open  smtp
80/tcp open  http
MAC Address: DE:AD:00:00:BE:EF (Intel Corporate)

# Nmap done at Tue Jun 16 12:15:03 2020 -- 1 IP address (1
host up) scanned in 10.22 seconds
```

Grepable Output

```
pecna@htb[/htb]$ cat target.gnmap# Nmap 7.80 scan initiated
Tue Jun 16 12:14:53 2020 as: nmap -p- -oA target 10.129.2.2
8Host: 10.129.2.28 ()   Status: Up
Host: 10.129.2.28 ()    Ports: 22/open/tcp//ssh///, 25/ope
n/tcp//smtp///, 80/open/tcp//http///    Ignored State: clos
ed (4)
```

```
# Nmap done at Tue Jun 16 12:14:53 2020 -- 1 IP address (1
host up) scanned in 10.22 seconds
```

XML Output

```
pecna@htb[/htb]$ cat target.xml<?xml version="1.0" encoding
="UTF-8"?>
<!DOCTYPE nmaprun>
<?xml-stylesheet href="file:///usr/local/bin/../share/nmap/
nmap.xsl" type="text/xsl"?>
<!-- Nmap 7.80 scan initiated Tue Jun 16 12:14:53 2020 as:
nmap -p- -oA target 10.129.2.28 -->
<nmaprun scanner="nmap" args="nmap -p- -oA target 10.129.2.
28" start="12145301719" startstr="Tue Jun 16 12:15:03 2020"
version="7.80" xmloutputversion="1.04">
<scaninfo type="syn" protocol="tcp" numservices="65535" ser
vices="1-65535"/>
<verbose level="0"/>
<debugging level="0"/>
<host starttime="12145301719" endtime="12150323493"><status
state="up" reason="arp-response" reason_ttl="0"/>
<address addr="10.129.2.28" addrtype="ipv4"/>
<address addr="DE:AD:00:00:BE:EF" addrtype="mac" vendor="In
tel Corporate"/>
<hostnames>
</hostnames>
<ports><extraports state="closed" count="4">
<extrareasons reason="resets" count="4"/>
</extraports>
<port protocol="tcp" portid="22"><state state="open" reason
="syn-ack" reason_ttl="64"/><service name="ssh" method="tab
le" conf="3"/></port>
<port protocol="tcp" portid="25"><state state="open" reason
="syn-ack" reason_ttl="64"/><service name="smtp" method="ta
ble" conf="3"/></port>
<port protocol="tcp" portid="80"><state state="open" reason
="syn-ack" reason_ttl="64"/><service name="http" method="ta
ble" conf="3"/></port>
```

```
</ports>
<times srtt="52614" rttvar="75640" to="355174"/>
</host>
<runstats><finished time="12150323493" timestr="Tue Jun 16
12:14:53 2020" elapsed="10.22" summary="Nmap done at Tue Ju
n 16 12:15:03 2020; 1 IP address (1 host up) scanned in 10.
22 seconds" exit="success"/><hosts up="1" down="0" total
="1"/>
</runstats>
</nmaprun>
```

Style sheets

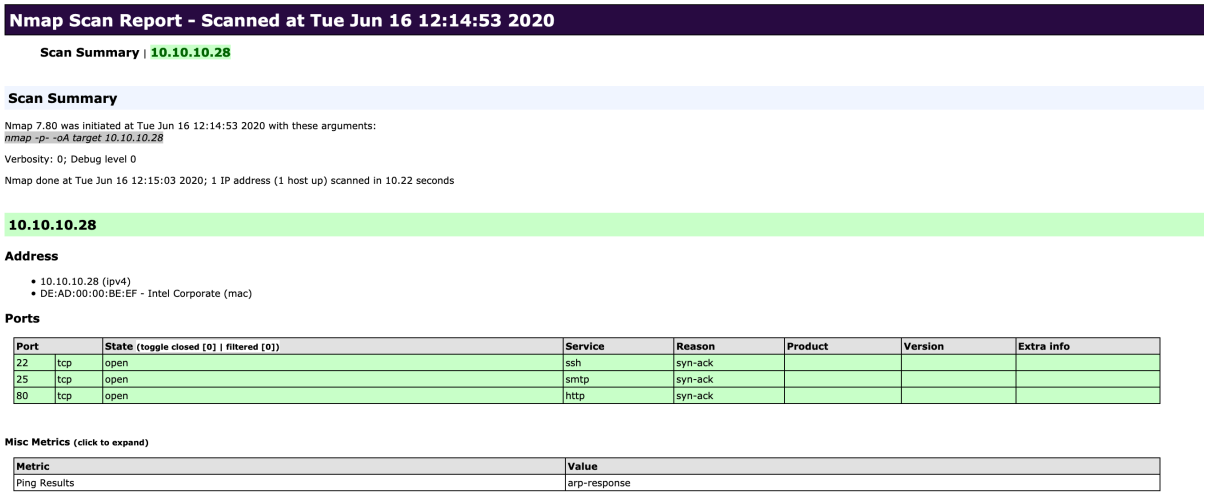
With the XML output, we can easily create HTML reports that are easy to read, even for non-technical people. This is later very useful for documentation, as it presents our results in a detailed and clear way. To convert the stored results from XML format to HTML, we can use the tool `xsltproc`.

Saving the Results

```
pecna@htb[/htb]$ xsltproc target.xml -o target.html
```

If we now open the HTML file in our browser, we see a clear and structured presentation of our results.

Nmap Report



Nmap Scan Report - Scanned at Tue Jun 16 12:14:53 2020

Scan Summary | **10.10.10.28**

Scan Summary

Nmap 7.80 was initiated at Tue Jun 16 12:14:53 2020 with these arguments:
nmap -p- -oA target 10.10.10.28

Verbosity: 0; Debug level 0

Nmap done at Tue Jun 16 12:15:03 2020; 1 IP address (1 host up) scanned in 10.22 seconds

10.10.10.28

Address

- 10.10.10.28 (ipv4)
- DE:AD:00:00:BE:EF - Intel Corporate (mac)

Ports

Port		State (toggle closed [0] | filtered [0])	Service	Reason	Product	Version	Extra info
22	tcp	open	ssh	syn-ack			
25	tcp	open	smtp	syn-ack			
80	tcp	open	http	syn-ack			

Misc Metrics (click to expand)

Metric	Value
Ping Results	arp-response